COMSATS University Islamabad, Lahore Campus

Terminal, Lab Examination – Spring 2025

Course Title:	Web Technologies			Course Code:	CSC336	Credit	3(2,1)
Course Instructor/s:	Asif Shahzad			Programme Name:	BSCS		
Semester:	6	Batch: FA22	Section:	G1	Date:	Jun. 19, 2025	
Time Allowed:	**2.5 Hours**			**Maximum Marks:**	**50**		
Student's Name:				Reg. No.			

Question 1: [50 Marks]

Create a web application with the following features. Use Bootstrap, Tailwind or custom CSS to decently style your pages. The app should function entirely from the homepage (`http://localhost:3000/`), means, without requiring user to enter any other URL in address bar. After each successful operation, redirect user to homepage.

(a) At "/" (i.e. homepage), show list of already added expenses (detailed below). Add a **Login** link in the header to open the login form. Show relevant error messages (e.g., invalid username/password) using sessions. After login, replace **Login** with a **Logout** link to log the user out. Valid username/password is **admin** / **1234**. [10 Marks]

(b) For logged in user, show: **Add Expense**, **Add Category** and **Show Report** links in header. When Add Expense is clicked, a form shall open to let user record expense **amount**, **description**, **date** and **category** from dropdown list. Save all expenses in **expenses.json** file. Expense categories shall be loaded from **categories.json** file e.g. Utility Bills, Grocery, Education, Traveling, etc. [10 Marks]

(c) In each expense row at home page, add Edit link to update expense, when clicked, open expense update form, where earlier expense data shall be pre-filled. Update expense in expenses.json file on submit. [10 Marks]

(d) Add Category shall allow user to add new expense category in categories.json file. [10 Marks]

(e) Show Report shall display total expense of each category in tabular form. [10 Mark]